Attention Multibranch Convolutional Neural Network for Hyperspectral Image Classification Based on Adaptive Region Search

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***Abstract*— Convolutional neural networks (CNNs) have demonstrated outstanding performance on image classification. To classify the hyperspectral images (HSIs), existing CNN-based approaches commonly adopt the architecture using single or several fixed spatial windows as inputs. This kind of architecture may lose contextual information or incorporate heterogeneous information due to the neglect of various land-cover distributions in HSIs. To deal with this problem, a novel attention multibranch CNN method based on adaptive region search (RS-AMCNN) is proposed for HSI classification. In RS-AMCNN, sizes and locations of spatial windows are searched in the nonlocal candidate region adaptively according to sample-specific distribution. These flexible spatial windows are input into several branches of RS-AMCNN. In each branch, convolutional long short-term memories (ConvLSTMs) are merged into CNN from shallow to deep layers, which not only extracts joint spatial–spectral features, but also exploits complementary information among different layers. Then, a branch attention mechanism is devised to emphasize more discriminative branches and suppress less useful ones. It forces RS-AMCNN to extract multiscale and multicontextual attention features for classification. Finally, RS-AMCNN is optimized end-to-end by combining the losses from the ramose classifiers of different branches and the main classifier. Experiments carried on several benchmark HSI data sets demonstrate that RS-AMCNN provides promising classification performance, especially in edge preservation and region uniformity.**

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***Index Terms*— Branch attention, convolutional long short-term memory (ConvLSTM), convolutional neural network (CNN), hyperspectral images (HSIs), region search.**

I. Introduction

RECENTLY, as the rapid development of optics and photonics, the hyperspectral imaging (HSI) techniques are significantly boosted, which can produce HSIs with higher resolution both in spectral and spatial dimensions. Hundreds of spectral bands are densely sampled ranging from ultraviolet to midinfrared spectrum, which provide HSIs with discriminative information to distinguish different objects [1]. Therefore, HSI processing plays an important role in various applications, including surveillance [2], geoponics [3], mineralogy [4], military [5], astronomy [6], and environmental sciences [7], [8].

HSI classification is the cornerstone of HSI processing [9]–[13]. The aim of HSI classification is to give every pixel in HSIs a corresponding class label. Feature extraction and classifier design are two key issues of the classification task. For feature extraction, a linear or nonlinear transformation is applied to map high-dimensional HSI data into a low-dimensional feature space [14]–[22]. At the beginning of HSI feature extraction, there are lots of linear-based methods, such as pairwise constraint discriminant analysis [16], principal component analysis [17], and balanced local discriminant embedding [18]. In general, HSI data are inherently nonlinear because of nonlinear factors in the hyperspectral imaging process [19]. Thus, many manifold learning-based methods are developed, such as sparse and low-rank near-isometric linear embedding [20] and semisupervised sparse manifold discriminative analysis [21]. These above-mentioned methods focus on spectral feature extraction of HSIs. When merely spectral information is involved, different land-cover classes cannot be distinguished precisely. Thus, a series of spatial–spectral techniques have been presented, for instance, 3-D Gabor filters [22] and 3-D morphological profiles [23]. These methods incorporate the spatial information to improve the classification performance of HSIs.

As for classifier design, representative methods include k-nearest neighbor [24], logistic regression [25], extreme learning machine [26], support vector machine (SVM) [27], and sparse representation-based classification [28]. Since SVM

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has obvious advantages in solving small sample size and high-dimensional problem, it has shown great potential in HSI classification.

Among the aforementioned methods, features are all handcrafted, which involve complicated feature engineering. Moreover, the training of feature extraction and classifier is separated. Recently, considerable literatures have grown up around the theme of deep learning methods. These deep learning methods have powerful automatic feature extraction capability, and are able to train feature extraction and classifier simultaneously. By stacking several neural network layers, deep learning models can extract hierarchical features and have been widely used in numerous applications, such as image classification [29] and object detection [30]. Inspired by these successful applications, different kinds of deep learning models have been developed for HSI classification. In [31], a deep stacked autoencoder (SAE) was proposed for HSI classification, which takes spectral signatures as the input to extract hierarchical features in an unsupervised manner. Later, a series of SAE variants were proposed, such as denoising SAE [32], sparse SAE [33], and Laplacian SAE [34]. In [35], a probabilistic generation model, deep belief network (DBN), was presented for HSI classification. These two kinds of deep learning models underuse spatial information because flattening operation causes the loss of spatial information. Besides, there are many full connection layers involving massive parameters.

Currently, there are several kinds of deep-learning-based methods for HSI classification. Recurrent neural networks (RNNs) are well known for its long-term data modeling ability [37] [55], [56], which are introduced to spectral feature extraction in HSIs. RNNs are usually heavily parameterized and a large number of training samples are required to guarantee the classification performance. Generative adversarial networks (GANs) ease the small sample size problem by generating high-quality samples [57]–[64]. GAN-based HSI classification methods mainly focus on semisupervised GANs [58], [59] and spatial–spectral GANs [61], [62]. Furthermore, graph convolutional networks (GCNs) model HSI data using graph structure, which alleviate this problem in a semi-supervised way [65], [66].

Inspired by visual perception of animals, CNN reduces the parameters by adopting local connections and sharing weights. It delivers the state-of-the-art results in HSI classification. Among CNN-based HSI classification methods, a large number of studies focus on joint spatial–spectral feature extraction of HSIs [36]–[39]. In [36], a dual-channel CNN (DCCNN) model was presented, where the spectral and spatial feature extractions are implemented by 1-D CNN and 2-D CNN, respectively. Furthermore, these two kinds of features are concatenated and fed into the softmax classification layer. Different from extracting spatial and spectral features separately, Chen *et al.* [37] presented 3DCNN method, where spectral and spatial features are extracted simultaneously. DCCNN and 3DCNN can extract local spectral information effectively using the convolution operation. To model the long-term dependencies among spectral bands, some advances on long-short-term memory (LSTM) model were proposed [38], [40]. In [38], a spectral–spatial unified network (SSUN) was developed to extract spatial features using multiscale CNN and spectral features using LSTM. Later, Zhou *et al.* [40] constructed the spectral LSTM and the spatial LSTM to learn spectral and spatial features of HSIs.

Apart from joint spatial–spectral feature extraction, many researchers focus on deeper model and sample augmentation. Generally speaking, a deeper neural network has better nonlinear representation ability. Thus, a series of deeper CNNs have emerged for HSI classification, such as contextual CNN [41], spatial–spectral residual network [42] and deep feature fusion network (DFFN) [43]. To ensure the performance of these deeper CNN models, sufficient training samples should be collected. However, the collection is time consuming and difficult in HSIs. In [44], a pixel-pair CNN (PPF-CNN) method was presented for HSI classification, which expands the training set by reorganizing the labeled pixels as the pixel pairs and assigning new labels to these pixel pairs. In [45], a semisupervised convolutional RNN was proposed. It pretrains the network using abundant unlabeled samples with their pseudo labels and then fine-tunes the network using limited labeled samples.

In these improved CNN-based methods [36]–[45], single fixed-size spatial window centered on a given sample is used as the input to maintain the spatial and spectral information of HSIs. This type of architecture may neglect various land-cover distributions of HSIs. Specifically, if the given center sample is located in the homogeneous regions, the surrounding samples generally have the same label with this sample. In this case, a larger spatial window is needed to contain as many surrounding samples as possible to provide enough contextual information for classification. In the heterogeneous regions, the given sample is generally close to the boundaries of different classes. In this case, some surrounding samples with different classes may be incorporated by enforcing the fixed center location for the given sample, which increases the difficulty of classification. To solve this problem, several researchers try to input multiple fixed spatial windows of different sizes and use multiple branches in CNNs. Feng *et al.* [46] proposed a divide-and-conquer dual-branch CNN (DDCNN) for HSI classification. It divides the HSIs into homogeneous and heterogeneous regions, then larger and smaller spatial windows are used in the homogeneous and heterogeneous regions accordingly. Zhang *et al.* [47] adopted central window, global window, and four direction-oriented spatial windows as the input of different branches in CNN. This method is abbreviated as DRCNN. Both DDCNN and DRCNN concatenate the features extracted from different branches for classification. DDCNN and DRCNN use several fixed spatial inputs and multiple branches to alleviate this problem to a certain extent. However, the fixed sizes and locations of spatial windows contradict various land-cover distributions of HSIs. Furthermore, features extracted from different branches have different discriminative abilities for classification. Treating all the branches equally may not be the best choice for classification.

In this article, a novel attention multibranch CNN method based on adaptive region search (RS-AMCNN) is proposed for HSI classification. In RS-AMCNN, several input spatial

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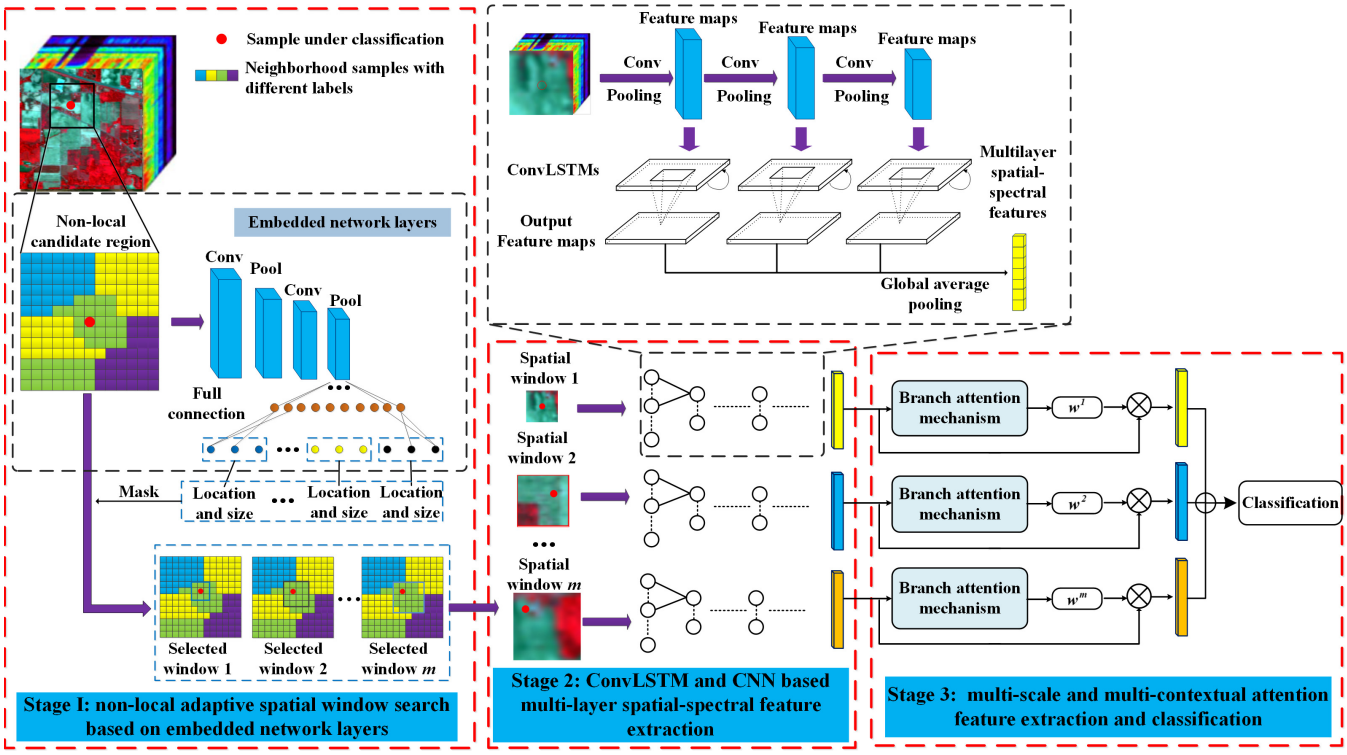


Fig. 1. Flowchart of the proposed RS-AMCNN method.

windows are searched in the nonlocal candidate region adaptively. Specifically, convolutional and full connection layers are constructed to obtain the location and size of each spatial window according to sample-specific spatial distribution. Then, these spatial windows are input to different branches of RS-AMCNN, respectively. In each branch, multilayer spatial–spectral features are extracted by constructing and adding convolutional LSTMs (ConvLSTMs) from shallow to deep layers. Thus, the softmax classification layer is added to each branch as the ramose classifier. To aggregate multiscale and multicontextual spatial–spectral features from different branches, a branch attention mechanism is designed. This mechanism forces RS-AMCNN to pay attention on more discriminative branches. It generates multiscale and multicontextual attention features by replying on predicted probabilities of ramose classifiers. Finally, RS-AMCNN is optimized using the loss from all the ramose classifiers and main classifier. The main contributions of this article can be summarized as follows.

1) To mitigate the contradiction between the fixed spatial inputs and various land-cover distributions of HSIs, an adaptive spatial window search strategy is designed and embedded in RS-AMCNN. It provides more potential for the improvement of classification performance by leveraging multiscale and multicontextual information.
2) To make full use of the characteristics of HSIs, joint spatial–spectral features are extracted by combining ConvLSTMs and CNN in each branch of RS-AMCNN. Furthermore, complementary information is obtained by fusing the abstract and detailed information among different layers.
3) In RS-AMCNN, branch attention mechanism adjusts the attention on different branches adaptively. Features from different branches are dynamically fused, which guarantees that the strong discriminative features contribute more to classification. Furthermore, RS-AMCNN combines multibranch feature extraction, branch attention selection, and classification into an end-to-end trainable network.

The remainder of this article is divided into three sections. The detailed contents of each section are introduced as follows. Section II introduces the architecture of the RS-AMCNN method thoroughly. Afterwards, the experimental results of RS-AMCNN and other state-of-the-art comparison methods are analyzed in Section III. At the end of this article, some conclusions and further work are provided in Section IV.

II. Proposed RS-AMCNN Method

RS-AMCNN mainly consists of three stages: nonlocal adaptive spatial window search based on embedded network layers, ConvLSTM and CNN-based multilayer spatial–spectral feature extraction, and multiscale and multicontextual attention feature extraction and classification. Fig. 1 shows the framework of the RS-AMCNN method. At the first stage, convolutional and full connection layers are embedded in RS-AMCNN to search the sizes and locations of spatial windows according to land-cover spatial distribution. These spatial windows are used as inputs of different branches of RS-AMCNN. At the next stage, ConvLSTMs are added to shallow, middle, and deep layers of each branch in RS-AMCNN, which extracts joint spatial–spectral features and makes full use of complementary information among different layers. At the final stage,

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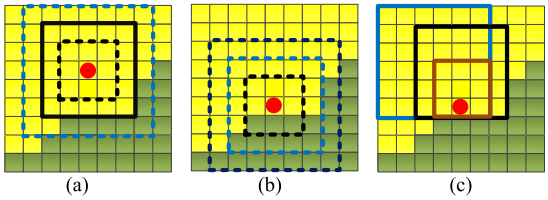


Fig. 2. Different selections of input spatial windows. (a) Larger spatial window for the sample in the homogenous region. (b) Smaller spatial window for the sample in the heterogeneous region. (c) Spatial windows with different sizes and positions for the sample in the heterogeneous region.

an attention branch mechanism allows each branch to adjust its weight adaptively. It dynamically fuses multibranch features for classification. RS-AMCNN is optimized using the loss from ramose and main classifiers. Finally, test samples are classified by the main classifier.

A. Nonlocal Adaptive Spatial Window Search Based on Embedded Network Layers

Most CNN-based HSI classification methods adopt single or several fixed spatial windows as the inputs to exploit spatial information. However, fixed input pattern greatly hinders the performance of CNNs for HSI classification. In HSIs, the spatial distribution of different samples is various. Thus, input spatial windows with adaptive locations and sizes are more suitable for HSI classification.

Fig. 2(a)–(c) shows different selections for input spatial windows. In Fig. 2, red circles indicate the samples to be classified, and squares in different colors indicate the samples with different classes. In Fig. 2(a), the sample to be classified is located in the homogenous region, and two spatial windows with different sizes are adopted. As shown in Fig. 2(a), a smaller spatial window marked by the dashed black line extracts insufficient contextual information for classification, whereas a larger spatial window marked by the dashed blue line incorporates some samples from different classes. In this case, the spatial window marked by the black solid line is a better choice. Thus, the adaptive sizes for spatial windows are necessary for classification. In Fig. 2(b), the classified sample is located in the heterogeneous region. As shown in Fig. 2(b), when the classified sample is in the center of spatial windows, no matter how to choose the sizes of spatial windows, there are some samples of different classes. In this case, it is necessary to adaptively adjust the position of the classified sample in the spatial window. In Fig. 2(c), three spatial windows with different sizes and locations are selected to represent the classified sample. Although there is some overlap between these spatial windows, these spatial windows can provide multiscale and multicontextual information. The changeable size and center location of spatial windows not only increase the flexibility of inputs, but also provide different levels of fine-grained information.

To deal with these situations, an adaptive spatial window search strategy is designed. It is embedded in RS-AMCNN to implement classification task-driven search. Given a sample, the locations and sizes of input spatial windows are searched in the nonlocal candidate region. In HSIs, it is more reasonable to utilize the neighborhood samples in the nonlocal region, which can judge the given sample by capturing rich contextual information. To embed the search into RS-AMCNN, the location and size of each input spatial window are encoded as a vector. The vector is obtained using the convolutional and full connection layers. To optimize the vector in the backward-propagation manner, the vector is transformed into a mask. Then, m spatial windows are generated using the mask in the nonlocal candidate region.

In the HSI data set, there are M labeled samples with d spectral bands $X = \{x_1, x_2, \ldots, x_M\}$ in an $R^{d\times 1}$ feature space. The corresponding labels are denoted by $Y = \{y_1, y_2, \ldots, y_M\}$. Given a sample $x_i (1 \le i \le M)$, Q_i indicates the set of neighborhood samples locating in the nonlocal candidate region centered on x_i. For all the labeled samples, a new data set $Q = \{Q_1, Q_2, \ldots, Q_M\}$ is obtained. It is divided into $Q^{\text{train}} = \{Q_1^{\text{train}}, Q_2^{\text{train}}, \ldots, Q_a^{\text{train}}\}$ and $Q^{\text{test}} = \{Q_1^{\text{test}}, Q_2^{\text{test}}, \ldots, Q_b^{\text{test}}\}$. $a+b = M$. In the nonlocal candidate region $Q-i$, m spatial windows need to be searched and N_i^q denotes the q th spatial window centered on x_i. For the q th spatial window, its size and location are encoded into a vector $T_i^q = [t_{x_i}^q, t_{y_i}^q, t_{l_i}^q]^T$, where $t_{x_i}^q$ and $t_{y_i}^q$ are horizontal and vertical coordinates of the center sample in the spatial window, and $t_{l_i}^q$ is the size of the spatial window.

In RS-AMCNN, convolutional layers and corresponding pooling layers are stacked to extract the hierarchical features of the nonlocal candidate region Q_i. Then, these features are mapped to a 1-D vector, which is taken as the input of the next full connection layer. Finally, T_i^q is obtained after the full connection layer. It is formulated as follows:

$$T_i^q = g[W_{\text{FC}} \cdot f(WQ_i + b) + b_{\text{FC}}] \tag{1}$$

where f is a composite function containing multiple convolutional and pooling operations, and W and b are the corresponding weight and bias. W_{FC} and b_{FC} are the weight and bias of the full connection layer, and $g(\cdot)$ denotes its activation function. Here, sigmoid is used to normalize the features between 0 and 1.

When the vector T_i^q is obtained, the corresponding spatial window can be cropped from the nonlocal candidate region. But the crop operation is difficult to be updated using the backward-propagation. Here, the boxcar function [48] is used to approximate the cropping operation. It transforms the vector T_i^q into a mask. Then, the spatial window N_i^q is achieved using element-wise multiplication between the nonlocal candidate region and the mask. It is formulated

$$N_i^q = Q_i \odot M\left(T_i^q\right) \tag{2}$$

where $M(\cdot)$ indicates the boxcar function and $\odot$ represents element-wise multiplication operation. Finally, the adaptive search of m spatial windows can be optimized through the final classification loss in the backward-propagation manner.

B. ConvLSTM and CNN Based Multilayer Spatial–Spectral Feature Extraction

HSIs have 3-D data structure, which have abundant information both in spatial and spectral dimensions. However, due to

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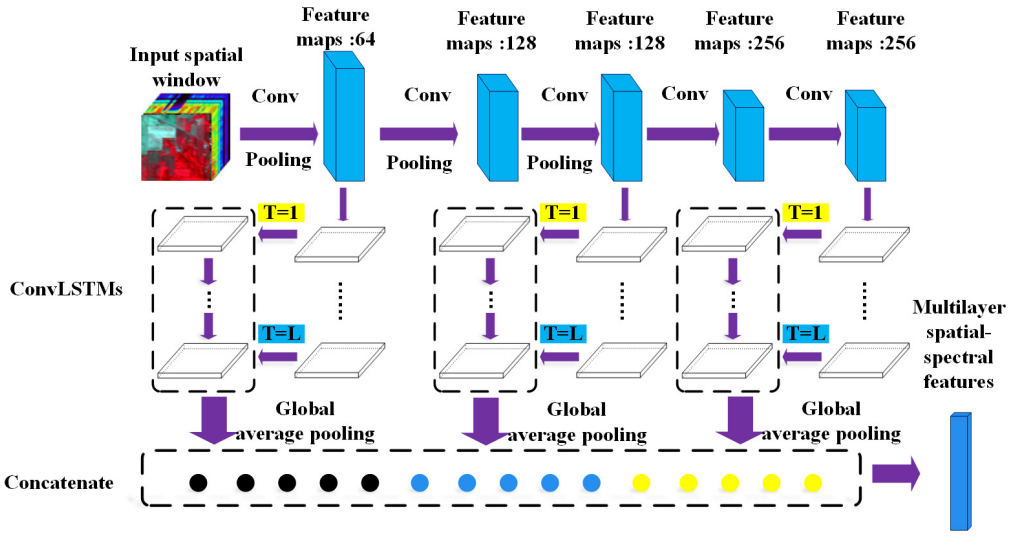


Fig. 3. Detailed architecture for one branch of RS-AMCNN.

the imaging mechanism of HSIs, some samples with different spectral signatures may correspond to the same class, whereas some similar samples in the spectral signature may have different labels. Thus, joint spatial–spectral feature extraction is significant for pixel-wise classification of HSIs. In addition, CNN-based methods extract hierarchical features from detail to abstraction, which provide different levels of discriminative information for classification. Thus, complementary information among different layers of CNNs is beneficial for classification.

In RS-AMCNN, multibranch CNN is constructed to extract features from searched spatial windows. Each spatial window is input to one branch of multibranch CNN. The detailed architecture for one branch of RS-AMCNN is shown in Fig. 3. In every branch of RS-AMCNN, convolutional and pooling layers are stacked successively to extract hierarchical features. These features are treated as spatio-spectral sequences, which are used as the inputs of ConvLSTMs. The following ConvLSTMs are merged into different convolutional layers from shallow to deep. In this case, shallow spatial–spectral features extract more detailed and edge information, whereas deep spatial–spectral features learn more abstract and semantic information. To make full use of coarse-to-fine information, multilayer spatial–spectral features are concentrated for classification.

LSTM is a special structure of the RNN. It shows outstanding performance on 1-D sequence data processing by learning the long-range dependency among adjacent or nonadjacent sequences. However, when dealing with 3-D spatiotemporal data, the full connection in LSTM fails to take rich spatial information into account. ConvLSTM [49] was proposed by replacing the full connection in LSTM with the convolution operation. It works with the memory cell c_t and three control gates (input gate i_t, forget gate f_t, and output gate o_t) based on convolution operation.

HSIs data are densely sampled from the visible to infrared spectrum. Since the spectral bands are approximately continuous, adjacent spectral bands have high correlation. Moreover, nonadjacent spectral bands may have long-term correlation. Thus, HSIs can be considered as the spectral sequence and the LSTM can be used to capture the long-term spectral dependencies through the recursion of the previous memory cell to the next memory cell.

To extract joint spatial–spectral information of HSIs, ConvLSTMs are merged into every branch of RS-AMCNN. Specifically, m spatial windows are input to corresponding m branches of RS-AMCNN. In the qth branch, the features extracted by convolutional layers from spatial windows N_i^q are represented as $F(N_i^q)$, which is regarded as 3-D spatio-spectral sequence. Then, $F(N_i^q)$ is divided into L subsequences$(F(N_i^q)^1, \ldots, F(N_i^q)^t, \ldots, F(N_i^q)^L)$ along the spectral dimension. These subsequences are input to ConvLSTMs through several successive recurrent cells. At the ith moment, memory cell c_t is calculated using the current input $F(N_i^q)^t$, the previous memory cell c_{t-1}, and hidden state h_{t-1} through the input gate i_t and forget gate f_t. Then, the hidden state h_t is computed by gathering information from current memory cell c_t and current input $F(N_i^q)^t$ through the output gate o_t. At the $t+1$th moment, the previous memory cell c_t and hidden state h_t are taken as the inputs to return a new memory cell c_{t+1}. The hidden state h_{t+1} of the $t+1$th moment is calculated in the same way as that of the tth moment. By this way, ConvLSTM establishes a long-term spectral relationship between subsequences using a recurrent form. At each moment, convolution operations of the input gate and forget gate are able to exploit spatial contextual information from the current input and previous hidden state. Thus, ConvLSTM learns the spectral dependencies and extracts spatial contextual features simultaneously.

To extract spatial–spectral features and make full use of complementary information among different layers, three ConvLSTMs are separately merged into shallow, middle, and deep layers of each branch in RS-AMCNN. In each branch, features extracted by three ConvLSTMs are input to the global average pooling layer, respectively. Then, these features are mapped into a 1-D vector. Here, the global average pooling operation

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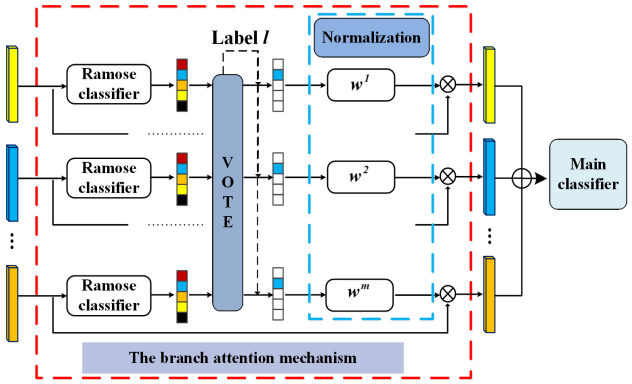


Fig. 4. Detailed architecture of the branch attention mechanism.

not only reduces numerous parameters, but also guarantees that RS-AMCNN adapts to the inputs with different scales. Finally, multilayer spatial–spectral representation is obtained by concatenating these vectors. This kind of representation learns complementary information of shallow appearance and deep semantic features. It is formulated as follows:

$$M_i^q = \text{Concat}\left[P\left(L_s\left(F\left(N_i^q\right)_s\right)\right), P\left(L_m\left(F\left(N_i^q\right)_m\right)\right), P\left(L_d\left(F\left(N_i^q\right)_d\right)\right)\right] \quad (3)$$

where $F(N_i^q)_s$, $F(N_i^q)_m$, and $F(N_i^q)_d$ denote the shallow, middle, and deep features extracted by convolutional layers in the q-th branch. $L_s(\cdot)$, $L_m(\cdot)$, and $L_d(\cdot)$ represent feature extracted by different ConvLSTMs, $P(\cdot)$ indicates global average pooling operation, Concat$(\cdot)$ is the concatenation operation and M_i^q represents multilayer spatial–spectral features of the qth branch for the ith sample.

C. Multiscale and Multicontextual Attention Feature Extraction and Classification

For multibranch network, how to fuse the features extracted from different branches is a key issue for classification. Most of multibranch networks [36]–[38] directly concatenate these features and feed them into the classifier. In this way, these features from different branches share the same weight. This kind of fusion is based on the assumption that features from each branch make equal contribution to classification. Actually, features from different branches usually have varied discriminative abilities for classification. It may not be the most discriminative feature set using the same weight for different branches. Thus, a more effective fusion strategy needs to be developed for multibranch classification.

To fuse the features from different branches adaptively, a branch attention mechanism is designed, which is shown in Fig. 4. In RS-AMCNN, different branches extract multiscale and multicontextual spatial–spectral features of HSIs. For a given sample in HSIs, some branches are more beneficial for classification because certain specific scale and contextual features are more discriminative than others. To take varied discriminative abilities of different branches into account, a branch attention mechanism emphasizes more discriminative branches and suppresses less useful ones. It evaluates the discriminative abilities of different branches by adding softmax layers as ramose classifiers. The ramose classifiers of different branches provide the predicted probabilities for the most possible class using voting strategy. Then, the branch attention mechanism assigns different weights for different branches according to their discriminative abilities adaptively. Features from different branches are weighted concatenated to generate multiscale and multicontextual attention feature representation for classification. Finally, these features are fed into the main classifier for classification.

In RS-AMCNN, adaptive weights are generated for all the branches and multiplied with the features extracted by corresponding branches. Then, these weighted features are concatenated to generate multiscale and multicontextual attention features. For the ith sample, the weighted features M_i are formulated as follows:

$$M_i = \text{Concat}\left(w_i^1 \cdot M_i^1, \ldots, w_i^q \cdot M_i^q, \ldots, w_i^m \cdot M_i^m\right) \quad (4)$$

where w_i^q denotes the weight of the qth branch, and M_i^q indicates multilayer spatial–spectral features extracted by the qth branch.

To obtain the weights adaptively, full connection and softmax layers are added to each branch as the ramose classifier. The softmax layer generates the predicted probabilities for all the classes. Take the qth branch as example, the full connection layer transforms the features M_i^q into a vector in an $R^{1\times C}$ feature space, where C represents the number of classes. Then, the predicted probability of the qth branch $p_{\text{rams}}^{\text{q}}(Q_i)$ is calculated using the softmax function on the feature vector. Similarly, for m branches, there are m predicted probabilities in total. These probabilities are denoted as $p_{\text{rams}}^{1}(Q_i), p_{\text{rams}}^{2}(Q_i), \ldots, p_{\text{rams}}^{\text{m}}(Q_i)$. In each branch, the class having the highest predicted probability is selected as the candidate class K_i^q. Among m branches, the most possible class is determined using majority voting strategy for these candidate classes. It is formulated as

$$K_i^q = \arg\max_{c=1,2,\ldots,C} \left[p_{\text{rams}}^{q}(Q_i)\right]_c, \quad q = 1, \ldots, m \quad (5)$$

$$l_i = \arg\max_{q=1,\ldots,m} \text{Count}\left(K_i^q\right) \quad (6)$$

where $[\cdot]_c$ denotes taking the cth value, $[p_{\text{rams}}^{\text{q}}(Q_i)]_c$ is the predicted probability of the cth class in $p_{\text{rams}}^{\text{q}}(Q_i)$, and the Count$(\cdot)$ is the function used for calculating the frequency of m candidate classes.

When the most possible class l_i is found for the ith sample, the predicted probabilities for the class l_i of all the branches are selected. The weight of each branch is obtained by normalizing these predicted probabilities. The weight of the qth branch is calculated as follows:

$$w_i^q = \frac{\left[p_{\text{rams}}^{\text{q}}(Q_i)\right]_{l_i}}{\sum_{j=1}^{m}\left[p_{\text{rams}}^{\text{j}}(Q_i)\right]_{l_i}} \quad (7)$$

where $[p_{\text{rams}}^{\text{q}}(Q_i)]_{l_i}$ is the predicted probability for the class l_i in the qth branch.

At the end of RS-AMCNN, a softmax layer is constructed as the main classifier to utilize the multiscale and multicontextual

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TABLE I
PROCEDURE OF THE PROPOSED RS-AMCNN METHOD

1. **INPUT:** The training data $Q_{train}=\{Q_1^{train},\ldots,Q_i^{train},\ldots,Q_a^{train}\}$ and the test data $Q_{test}=\{Q_1^{test},Q_2^{test},\ldots,Q_b^{test}\}$ from C classes, the class labels of training samples $Y_{train}=\{y_1,y_2,\ldots,y_a\}$, the mini-batch size and the number of training epochs E
2. **Begin**
3. randomly initialize weight matrices and biases of RS-AMCNN
4. input the training data Q_{train} with all the spectral bands into RS-AMCNN
5. **for** E epochs
6. **for** training samples of every mini-batch
7. compute the unified loss function from main and ramose classifiers by (8)
8. update the parameters of RS-AMCNN by minimizing the loss (8)
9. **end for**
10. **end for**
11. calculate the labels of test data $Q_{test}=\{Q_1^{test},Q_2^{test},\ldots,Q_b^{test}\}$
12. **END**
13. **OUTPUT:** the labels of the test samples Q_{test} are obtained by the trained RS-AMCNN

attention features M_i for final classification. In the training process, the loss of RS-AMCNN is defined as

$$J(\theta)=-\frac{1}{a}\left(\sum_{i=1}^{a}\sum_{c=1}^{C}I\{c=y_i\}\log\left(p_{\text{main}}\left(Q_{\text{i}}^{\text{train}}\right)\right)+\sum_{q=1}^{m}\sum_{i=1}^{a}\sum_{c=1}^{C}I\{c=y_i\}\log\left(p_{\text{rams}}^{\text{q}}\left(Q_{\text{i}}^{\text{train}}\right)\right)\right) \quad (8)$$

where $p_{\text{main}}(Q_{\text{i}}^{\text{train}})$ represents the predicted probability of the main classifier for the ith training sample Q_i^{train}. The first term of (8) is the loss of the main classifier, which is calculated by the cross entropy of the real-class probability and the output class probability. The second term of (8) is the losses of m ramose classifiers calculated in the same way.

In the test stage of RS-AMCNN, the label of the ith testing sample $Q_{\text{i}}^{\text{test}}$ is predicted by computing the maximum probability of different classes, which is calculated by

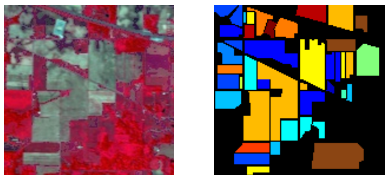
$$\text{Class}\left(Q_{\text{i}}^{\text{test}}\right)=\arg\max_{c=1,2,\ldots,C}\left[p_{\text{main}}\left(Q_{\text{i}}^{\text{test}}\right)\right]_c \quad (9)$$

where $[p_{\text{main}}(Q_{\text{i}}^{\text{test}})]_c$ represents the predicted probability for the cth class from the main classifier.

RS-AMCNN combines nonlocal adaptive spatial window search, multiscale, and multicontextual attention feature extraction and classification into a unified framework. The higher discriminative ability the branch has, the larger the weight of the branch is. It guarantees the branch having higher discriminative ability contributes more to the classification.

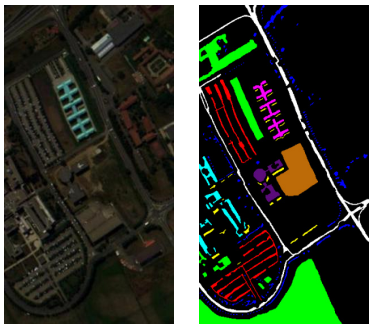
D. Procedure of RS-AMCNN

The proposed RS-AMCNN method combines multibranch feature extraction, branch selection, and classification into a unified optimization procedure. To describe the RS-AMCNN clearly, the procedure of the RS-AMCNN method is provided in Table I.



(a) (b)

Fig. 5. Indian Pines data set. (a) False-color composite image. (b) Ground truth.



(a) (b)

Fig. 6. Pavia University data set. (a) False-color composite image. (b) Ground truth.

III. EXPERIMENTAL RESULTS

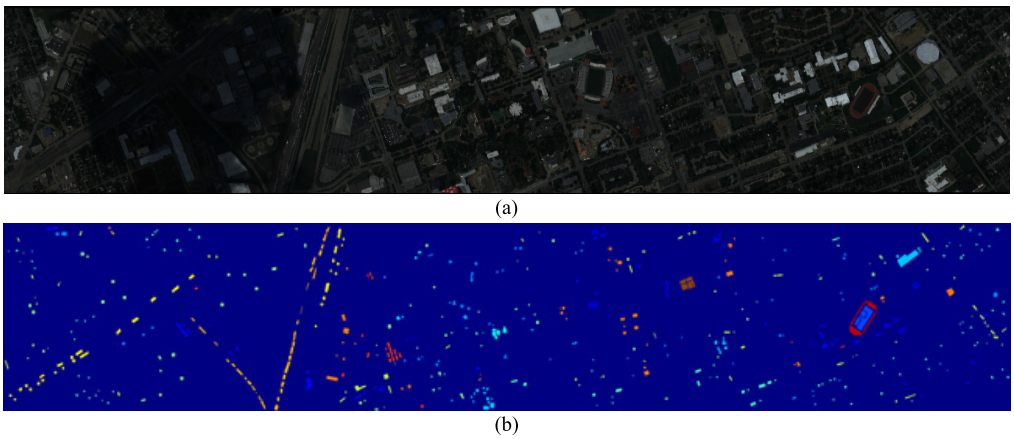
In this section, the performance of the newly developed RS-AMCNN method is evaluated based on three open benchmark hyperspectral data sets. To valid the effectiveness of RS-AMCNN, eight state-of-the-art comparison methods are used. The RS-AMCNN method is evaluated from various aspects, including classification performance, running time, sensitivity analysis to the number of training samples, effectiveness analysis of each step and sensitivity analysis of free parameters, and effectiveness analysis of joint spatial–spectral feature extraction in RS-AMCNN.

A. Data Description

The proposed RS-AMCNN method is conducted on three representative HSI data sets: the Indian Pines data set, the Pavia University data set, and the University of Houston data set. The corresponding false-color composite images and ground-truths for three data sets are shown in Figs. 5–7. More detailed introductions of three data sets are provided as follows.

1) The Indian Pines data set was captured by the AVIRIS over an agriculture area in June 1992. The image is low spatial resolution with a spatial size of 145 × 145. After removing noisy spectral bands, 200 spectral bands have been selected from original 224 spectral bands ranging from 200to 2400 nm. There are 16 vegetation categories to be classified.
2) The Pavia University data set was acquired by the ROSIS-3 sensor in an urban site over Pavia, northern Italy. The image has higher spatial resolution than Indian Pines, which covers an area with 610 × 345 pixels and has the spatial resolution of a 1.3 m per pixel. There are totally 103 bands available after discarding 20 water

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(a) (b)

Fig. 7. University of Houston data set. (a) False-color composite image. (b) Ground truth.

TABLE II
DETAILED STRUCTURE OF RS-AMCNN

Part	No	Operation	Activation	ConvLSTM?
Spatial window search	1	Conv: 3×3×32 + Pooling: 2×2	ReLu	
	2	Conv: 3×3×16 + Pooling: 2×2	ReLu	
	3	Fully connected	Sigmoid	
Feature extraction for every branch	1	Conv: 3×3×64 + Pooling: 2×2	ReLu	Yes
	2	Conv: 3×3×128 + Pooling: 2×2	ReLu	No
	3	Conv: 3×3×128 + Pooling: 2×2	ReLu	Yes
	4	Conv: 3×3×256	ReLu	No
	5	Conv: 3×3×256	ReLu	Yes
	Add to layer 1	ConvLSTM: 3×3×32 + GAP	Tanh/Sigmoid	
	Add to layer 3	ConvLSTM: 3×3×64 + GAP	Tanh/Sigmoid	
	Add to layer 5	ConvLSTM: 2×2×128 + GAP	Tanh/Sigmoid	
Classification	1	Fully connected	Softmax	

absorption or corrupted bands. The ground-truth differentiates nine land-cover classes.

3) The University of Houston data set was collected by the NSF-funded Center over the University of Houston campus and its adjacent downtown area. This data set is characterized by high spatial resolution with 349 × 1905 pixels. It consists of 144 spectral bands ranging from 364 to 1046 nm. There are totally 15 different categories.

B. Experimental Setting

To verify the effectiveness of the RS-AMCNN method, eight HSI classification methods are used for comparison. Most of the methods are state-of-the-art deep-learning-based methods, including SAE [31], DBN [35], CNN [50], PPF-CNN [44], 3DCNN [37], DFFN [41], and DR-CNN [47]. In addition, a classical method, SVM with radial basis function (RBF-SVM), is also used [27]. To provide a quantitative evaluation, overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa) are utilized as the measurements of classification performance. All experiments are executed for over 20 times independently with randomly divided training and test sets. These experiments are carried out using Python language and Pytorch library on a NVIDIA 2080Ti graphics card.

For RBF-SVM, one-against-all strategy is used to achieve multiclassification and five-fold cross validation is adopted to determine the penalty and gamma parameters. For SAE and DBN, the size of the spatial window is searched ranging from 3 to 21 with an interval of 2. For CNN, 5 × 5 spatial window is adopted as suggested in [37]. For PPF-CNN, the size of voting window is set to the default value in [44]. For 3DCNN, the spatial window size of 3-D input is resized to 27 × 27 × 100 [37]. For DR-CNN, the sizes of central spatial window and original spatial window are set as 3 × 3 and 11 × 11. The sizes of four direction-oriented spatial windows are set to the half size of original spatial window along different directions. For DFFN, the numbers of convolutional layers are set to be 27, 33, and 27 for Indian Pines, Pavia University, and University of Houston, respectively [43].

The main structure and parameters of RS-AMCNN are shown in Table II. It records the experimental settings of

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TABLE III
NUMBERS OF TRAINING AND TEST SAMPLES PER CLASS ON THE INDIAN PINES DATA SET

Class		Numbers of samples	
No	Name	Training	Test
1	Alfalfa	2	44
2	Corn-notill	71	1357
3	Corn-mintill	42	788
4	Corn	12	225
5	Grass-pasture	24	459
6	Grass-trees	36	694
7	Grass-pasture-mowed	1	27
8	Hay-windrowed	24	454
9	Oats	1	19
10	Soybean-notill	49	923
11	Soybean-mintill	123	2332
12	Soybean-clean	30	563
13	Wheat	10	195
14	Woods	63	1202
15	Buildings-Grass-Trees-Drives	19	367
16	Stone-Steel-Towers	5	88
	Total	512	9737

nonlocal adaptive spatial window search, feature extraction for every branch, and classification. In RS-AMCNN, the number of branches is set to be five empirically. RS-AMCNN uses a mini-batch gradient descent for training, where the size of mini-batch is 128, the initial learning rate is 0.002, and the number of epochs is 400. These parameters are determined by a trial-and-error procedure.

C. Classification Results of Hyperspectral Data Sets

1) Classification Results of the Indian Pines Data Set: The training set consists of 5% of the samples in the data set and the test set is made up of the remaining 95%. The specific number of samples per class for training and test is summarized in Table III.

Table IV records the average classification accuracies and the standard deviations of the nine methods over 20 executions. The first 16 rows record the class-specific classification accuracy and the last three rows record the OA, AA, and Kappa of all the classes. Among nine algorithms, the best results are highlighted by gray regions. As shown in Table IV, deep-learning-based methods, SAE, DBN, CNN, PPF-CNN, 3DCNN, DFFN, DR-CNN, and RS-AMCNN outperform RBF-SVM because of the excellent nonlinear and hierarchical feature extraction ability. PPF-CNN outperforms CNN by leveraging a tremendous number of available training samples. Compared with PPF-CNN and 3DCNN, DFFN designs deeper architecture to improve the classification performance. DR-CNN performs better than DFFN due to the usage of multiple input spatial windows. Compared with DR-CNN, RS-AMCNN is further improved using adaptive spatial window search and multilayer spatial–spectral feature extraction. Among the nine methods, RS-AMCNN obtains the highest classification accuracies not only in most classes but also in the OA, AA, Kappa indexes. It is noticeable that RS-AMCNN provides completely correct prediction for some classes, such as the corn, Hay-windrowed, and oats. On the whole, RS-AMCNN improves by at least 3.7% in OA, 8.5% in AA, and 4.3% in Kappa.

Fig. 8(a)–(i) visualizes the classification results of the nine methods. RBF-SVM, SAE, DBN, and PPF-CNN generate the classification maps with many noisy scattered points, especially in the corn-noill, corn-mintill. Soybean-notill, and soybean-mintill classes. 3DCNN incorporates spatial information for classification and significantly reduces the scattered points. Compared with other methods, RS-AMCNN greatly improves regional consistency and boundary localization of some classes, such as soybean-notill and Corn-notill.

2) Classification Results of the Pavia University Data Set: The training set consists of 3% of the samples in the data set and the test set is made up of the remaining 97%. The specific number of samples per class for training and test samples is summarized in Table V.

For the Pavia University data set, the quantitative classification results are summarized in Table VI. As reported in Table VI, 3DCNN, DFFN, and DR-CNN are more excellent than RBF-SVM, SAE, and DBN because the convolution operation captures locally-invariant spatial features. For the gravel class, RBF-SVM, SAE, DBN, and CNN achieve unsatisfactory classification results. Compared with these methods, RS-AMCNN improves by 39.2%, 26.3%, 28.7%, and 20.5%, respectively. For all the classes, RS-AMCNN achieves over 95% classification accuracies. Particularly, RS-AMCNN achieves absolutely correct classification results in the meadows and bare soil classes. In terms of OA, AA, and Kappa indexes, RS-AMCNN obtains the best classification result among all the nine methods.

The corresponding visual classification maps provide a more intuitive comparison between nine methods in Fig. 9(a)–(i). RBF-SVM, SAE, DBN, and PPF-CNN misclassify a certain number of samples of the bitumen class as the asphalt class. It is difficult to distinguish the samples of bitumen and asphalt classes with similar spectral signature. For 3DCNN, DFFN, and DR-CNN, some samples belonging to the bare soil class are misclassified as the meadows class. In case of these two situations, RS-AMCNN provides more distinctions between different classes.

3) Classification Results of the University of Houston Data Set: About 5% samples from each class are selected for training and the other 95% are used for test. The detailed number of 15 classes of samples is summarized in Table VII.

The statistical classification results on the University of Houston data set are recorded in Table VIII. It can be seen that spectral-based methods, RBF-SVM, SAE, and DBN, achieve similar classification results. All these methods misclassify many samples in the parking_lot2 class. Compared with these methods, other CNN-based methods significantly improve the classification performance of the parking_lot2 class, especially RS-AMCNN. RS-AMCNN has 62.3%, 58.1%, and 60.3% improvement, respectively. RS-AMCNN achieves higher classification accuracies in most classes than other methods, especially in the grass_stressed, tennis_court, and runing_track classes. In addition, RS-AMCNN increases by

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TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE INDIAN PINES DATA SET

Class	RBF-SVM [25]	SAE[29]	DBN[33]	CNN[47]	PPF-CNN [41]	3DCNN[35]	DFFN[40]	DR-CNN [44]	RS-AMCNN
1	6.1±11.2	10.0±6.4	13.6±5.6	78.4±10.2	50.4±8.4	83.8±13.4	88.6±10.9	95.3±3.7	98.7±0.9
2	72.9±3.6	79.7±2.3	79.8±2.9	75.4±2.4	89.2±2.1	92.7±3.5	87.5±2.8	87.6±4.7	98.3±0.6
3	58.0±3.6	74.9±4.8	70.5±2.2	82.8±3.3	77.1±2.7	87.2±10.4	89.2±2.9	91.6±5.3	98.5±0.6
4	39.0±15.0	62.8±8.3	71.3±6.6	89.2±3.5	87.7±3.7	83.4±8.3	91.6±7.7	88.2±3.5	100.0±0.0
5	87.0±4.5	84.2±3.3	80.1±4.1	69.0±4.6	92.7±1.0	84.0±5.7	92.2±2.4	85.0±2.9	98.0±0.4
6	92.4±2.0	94.3±1.7	94.2±2.4	92.8±2.5	93.1±1.9	93.4±2.5	98.1±3.3	97.9±3.2	99.6±0.0
7	0±0	24.4±18.8	28.1±22.6	51.1±12.3	0±0	97.2±4.8	51.9±16.1	92.3±2.0	95.0±0.0
8	98.1±1.4	98.8±0.4	98.5±1.5	97.1±1.6	99.6±0.3	97.4±2.8	98.2±3.7	99.1±2.6	100.0±0.0
9	0±0	11.1±10.1	9.5±2.4	41.6±9.9	0±0	77.0±11.1	57.9±19.6	61.6±9.0	100.0±0.0
10	65.8±3.7	73.6±3.8	73.2±4.7	81.0±2.6	85.6±2.8	93.3±5.0	90.8±3.3	94.3±3.6	99.3±0.2
11	85.3±2.9	83.4±2.0	82.7±2.2	87.2±1.5	83.8±1.6	94.9±2.7	96.6±1.6	97.7±2.7	99.8±0.1
12	69.6±6.5	70.4±8.0	62.0±5.8	84.4±2.3	90.4±3.1	89.8±4.3	98.6±4.1	92.7±3.2	99.1±0.8
13	92.3±4.1	94.2±4.3	89.7±10.6	83.1±4.2	97.8±0.9	92.8±5.9	96.9±3.3	98.0±0.9	100.0±0.0
14	96.6±1.0	94.2±1.5	94.4±1.6	98.2±0.8	95.5±1.1	98.3±1.3	98.1±1.6	99.6±0.3	98.4±0.4
15	41.7±7.0	66.1±5.6	64.2±6.5	84.7±4.5	78.0±2.4	77.8±13.4	86.6±5.3	98.3±1.2	98.4±0.6
16	75.2±9.0	87.6±8.1	80.5±13.2	76.0±8.1	97.3±1.3	88.4±5.3	72.7±11.2	76.0±5.5	96.5±2.1
OA（%）	77.8±0.8	81.9±0.1	80.6±0.1	85.4±0.8	87.9±0.8	92.8±0.8	93.5±1.2	95.0±0.9	98.7±0.5
AA（%）	61.3±1.4	69.4±1.9	68.3±1.7	79.4±1.6	76.5±0.6	89.4±1.4	87.2±2.6	90.2±1.9	98.7±0.7
Kappa(%)	74.5±1.0	79.3±1.1	77.8±1.3	84.3±2.9	86.3±0.9	91.9±0.9	92.6±1.4	94.3±1.1	98.6±0.4

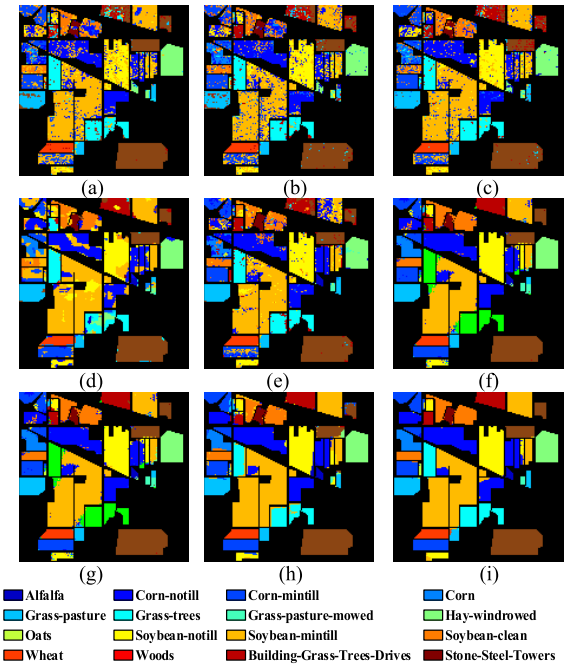


Fig. 8. Classification visual maps of all the nine methods on the Indian Pines data set. (a) RBF-SVM. (b) SAE. (c) DBN. (d) CNN. (e) PPF-CNN. (f) 3DCNN. (g) DFFN. (h) DR-CNN. (i) RS-AMCNN.

at least 2.9%, 2.4%, and 3.1% in the OA, AA, and Kappa indexes.

Fig. 10(a)–(i) shows the classification visual maps of the nine algorithms. For RBF-SVM, SAE, and DBN, there are many noisy scatter points in the classification maps. DFFN and DR-CNN greatly improve the region uniformity in many classes, such as the residential and road classes. Compared with DFFN and DR-CNN, RS-AMCNN is better at keeping boundary in the running_track class and achieves better region uniformity in the commercial class.

TABLE V
NUMBERS OF TRAINING AND TEST SAMPLES PER CLASS ON THE PAVIA UNIVERSITY DATA SET

Class		Numbers of samples	
No	Name	Training	Test
1	Asphalt	199	6432
2	Meadows	559	18090
3	Gravel	63	2036
4	Trees	92	2972
5	Painted metal sheets	40	1305
6	Bare Soil	151	4878
7	Bitumen	40	1290
8	Self-Blocking Bricks	110	3572
9	Shadows	28	919
Total		1282	41494

D. Investigation on Running Time

Tables IX–XI show the execution time in the training and test process for all the algorithms on three data sets, respectively. Compared with RBF-SVM, other deep-learning-based methods require more time to train the networks. The reason why DBN and SAE are faster than other deep-learning-based methods is that their inputs are 1-D spectral signatures with less computational burden. Among all the methods, PPF-CNN and 3DCNN are the most time consuming. For the former, it costs more training time because a lot of training samples are generated by the data expansion. For the latter, a large number of parameters in 3-D convolutional kernels result in long training time. DR-CNN costs a relatively more time because of its multi-branch structure. RS-AMCNN costs more training time than DFFN, but less than PPF-CNN, 3DCNN, and DR-CNN. For the test time, RS-AMCNN and DR-CNN take more than SAE, DBN, CNN, and DFFN because of multibranch CNN architecture.

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TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATA SET

Class	RBF-SVM [25]	SAE[29]	DBN[33]	CNN[47]	PPF-CNN [41]	3DCNN[35]	DFFN[40]	DR-CNN [44]	RS-AMCNN
1	90.7±1.1	92.3±1.1	91.6±0.8	93.1±1.4	98.0±0.1	95.5±1.2	94.9±1.8	98.3±0.8	99.7±0.1
2	96.8±0.7	97.6±0.3	97.4±0.4	97.6±0.9	99.2±0.2	99.4±0..3	98.6±0.4	99.6±0.2	100.0±0.0
3	60.2±5.4	72.1±3.5	69.7±6.0	77.9±4.5	84.9±1.8	92.6±5.4	93.5±2.9	93.9±2.8	98.4±0.1
4	90.8±2.0	90.9±1.4	91.2±1.4	86.4±3.6	95.8±0.8	78.2±4.9	90.2±3.0	98.3±0.3	98.2±0.2
5	98.8±0.4	98.7±0.4	98.6±0.6	98.5±1.4	99.8±0.1	95.4±4.3	92.6±2.8	99.9±0.2	99.9±0.1
6	79.5±4.9	86.9±1.9	85.6±2.2	91.0±2.8	96.4±0.3	96.4±0.6	96.5±1.2	97.8±2.0	100.0±0.0
7	74.3±5.1	78.1±4.9	74.8±4.8	81.2±2.9	89.2±0.8	91.5±3.4	93.9±4.8	93.7±2.9	99.4±0.1
8	88.8±2.2	87.8±1.4	88.2±1.3	92.5±2.2	93.7±1.2	94.8±1.4	96.4±1.1	97.1±0.9	99.0±0.1
9	99.8±0.1	99.5±0.3	99.6±0.1	79.0±4.1	98.5±0.7	77.4±2.8	90.7±0.8	98.6±0.2	95.4±0.4
OA（%）	90.3±0.6	92.4±0.3	91.9±0.3	93.0±0.6	96.9±0.2	95.2±0.7	96.1±0.4	98.1±0.2	99.5±0.1
AA（%）	86.6±0.9	89.3±0.7	88.5±0.8	88.6±0.8	95.1±0.2	91.2±1.1	94.1±0.3	97.0±0.4	98.8±0.1
Kappa(%)	87.1±0.8	89.9±0.4	89.2±0.4	90.7±0.7	96.0±0.2	93.8±0.9	95.0±0.6	97.5±0.2	99.3±0.1

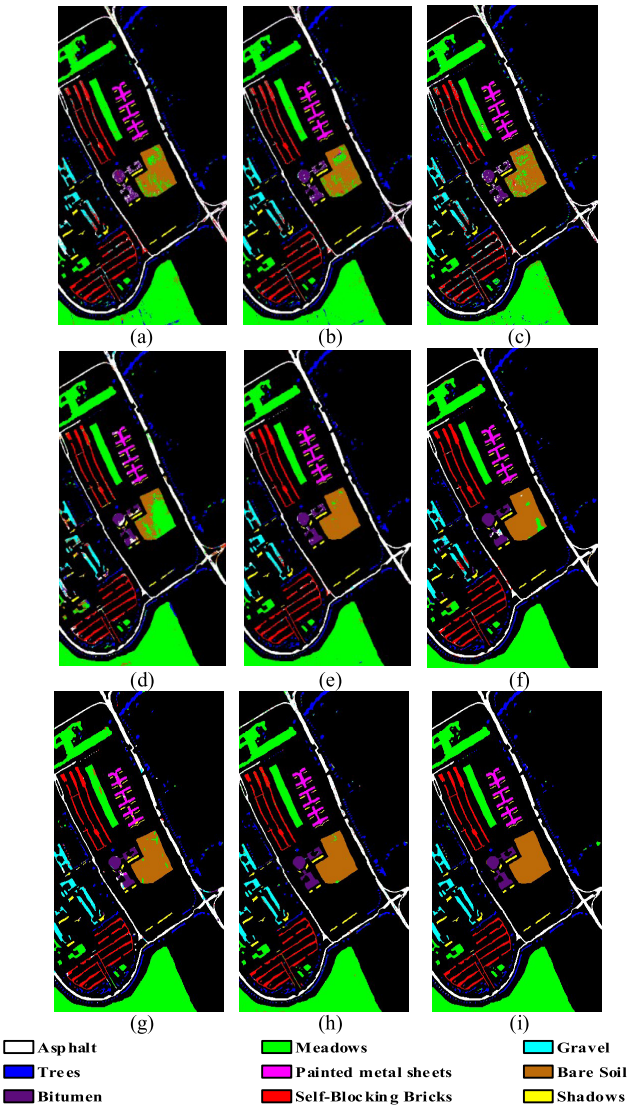


Fig. 9. Classification visual maps of all the nine methods on the Pavia University data set. (a) RBF-SVM. (b) SAE. (c) DBN. (d)CNN. (e) PPF-CNN. (f) 3DCNN. (g) DFFN. (h) DR-CNN. (i) RS-AMCNN.

E. Sensitivity to the Number of Training Samples

In Fig. 11(a)–(c), all the methods are investigated with different numbers of training samples and the corresponding

TABLE VII
NUMBERS OF TRAINING AND TEST SAMPLES PER CLASS ON THE UNIVERSITY OF HOUSTON DATA SET

Class		Numbers of samples	
No	Name	Training	Test
1	grass_healthy	63	1188
2	grass_stressed	63	1191
3	grass_synthetic	35	662
4	tree	62	1182
5	soil	62	1180
6	water	16	309
7	residential	63	1205
8	commercial	62	1182
9	road	63	1189
10	highway	61	1166
11	railway	62	1173
12	parking_lot1	62	1171
13	parking_lot2	23	446
14	tennis_court	21	407
15	running_track	33	627
Total		751	14278

OA results are presented in a histogram form. Specifically, 1%, 3%, 5%, 7%, 9% samples on the Indian Pines data set, 1%, 2%, 3%, 4%, 5% samples on the Pavia University data set, and 1%, 3%, 5%, 7%, 9% samples on the University of Houston data set per class are randomly selected from the labeled samples. When 5% samples for the Indian Pines, 3% samples for the Pavia University, and 5% samples for the University of Houston are used, CNN, DFFN, DR-CNN, and RS-AMCNN have obvious advantages over other methods, especially the proposed RS-AMCNN

method. When the number of training samples decreases, it is obvious that the OA values of all the methods decline. In this case, compared with RBF-SVM, some deep-learning-based methods, such as SAE, DBN, and CNN, have no obvious advantages. PPF-CNN performs well using the expansion of training set, whereas 3DCNN and CNN are not well trained with fewer training samples. When only 1% of the training samples are used on three data sets, the classification performance of DFFN deteriorates quickly because it is heavily parameterized. Compared with other methods, RS-AMCNN shows better classification performance with small-sized training sample set by leveraging multiscale and multicontextual

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TABLE VIII
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE UNIVERSITY OF HOUSTON DATA SET

Class	RBF-SVM [25]	SAE[29]	DBN[33]	CNN[47]	PPF-CNN [41]	3DCNN[35]	DFFN[40]	DR-CNN [44]	RS-AMCNN
1	96.7±1.1	95.4±2.6	97.1±2.3	94.1±0.2	96.8±2.2	92.7±2.0	98.3±1.4	99.2±1.4	97.8±0.5
2	97.7±1.0	97.6±1.4	97.5±1.2	95.1±0.6	97.0±1.0	93.4±2.2	98.0±1.0	97.3±1.0	100.0±0.0
3	99.7±0.1	98.1±0.5	96.0±0.4	98.9±0.5	99.9±0.1	99.2±0.6	99.7±0.2	99.8±0.2	98.0±0.6
4	96.3±2.5	94.7±2.2	96.9±1.1	87.8±4.4	95.0±2.8	87.8±2.4	96.5±1.8	95.2±1.8	99.6±1.5
5	98.0±0.6	97.4±1.7	96.9±1.2	98.9±0.3	99.6±0.3	97.5±3.4	99.4±0.4	99.2±0.4	99.8±0.1
6	82.9±5.8	93.6±0.6	92.2±0.3	95.5±4.6	85.7±1.8	86.6±6.3	76.3±7.6	92.7±7.6	96.9±1.7
7	86.8±1.9	86.9±2.1	83.2±2.4	93.2±2.4	92.9±3.0	91.8±3.3	94.7±1.3	94.5±1.3	96.2±0.4
8	81.1±3.4	79.0±2.1	80.4±1.6	75.2±5.3	86.4±1.0	91.5±0.8	88.5±3.5	91.3±3.5	97.0±0.8
9	81.1±3.6	79.3±4.7	77.1±3.9	87.7±3.0	86.2±0.6	91.5±2.2	89.3±4.8	91.7±4.8	98.3±0.2
10	89.3±3.6	86.5±2.1	87.4±1.9	92.9±1.6	91.0±1.7	91.9±4.8	91.1±2.9	98.0±2.9	99.6±0.0
11	80.4±4.6	82.0±0.2	84.2±0.1	97.9±1.0	91.5±1.2	96.5±2.0	86.7±2.7	97.5±2.7	99.8±0.2
12	76.8±5.9	82.0±0.9	81.8±0.5	89.1±2.3	89.9±1.9	93.6±2.3	94.8±1.1	91.5±1.1	98.6±0.4
13	32.5±4.6	36.7±2.3	34.5±2.2	75.1±4.7	71.2±3.4	83.0±9.8	83.1±1.1	91.3±1.1	94.8±0.2
14	97.1±1.8	97.9±1.4	98.6±0.9	96.7±1.7	100.0±0.0	99.9±0.2	93.5±3.3	100.0±0.0	100.0±0.0
15	98.4±0.2	99.2±0.3	98.7±0.2	98.1±1.8	98.4±0.4	99.4±0.6	99.9±0.1	99.5±0.1	100.0±0.0
OA（%）	87.7±1.1	87.8±0.7	87.3±0.9	91.6±0.9	92.6±0.4	93.2±0.8	93.6±0.5	95.9±0.5	98.7±0.2
AA（%）	86.3±0.6	87.1±0.6	86.8±0.5	30.6±0.4	92.1±0.4	93.1±1.2	92.6±0.6	95.9±0.6	98.3±0.3
Kappa(%)	86.8±0.6	86.9±0.7	86.2±0.6	90.9±1.0	92.0±0.4	92.6±0.9	93.0±0.5	95.5±0.5	98.6±0.2

TABLE IX
RUNNING TIME OF DIFFERENT METHODS ON THE INDIAN PINES DATA SET

Dataset	Method	Training Time (s)	Test Time (s)
Indian Pines	RBF-SVM[27]	0.4±0.1	1.2±0.1
	SAE[32]	76.3±8.4	0.2±0.1
	DBN[35]	114.3±20.1	0.2±0.1
	CNN[50]	220.7±27.9	0.5±0.1
	PPF-CNN[44]	2056.0±36.7	5.3±0.3
	3DCNN[37]	2690.2±57.9	16.0±0.1
	DFFN[43]	116.4±0.3	3.8±0.2
	DR-CNN[47]	1450.2±19.6	102.8±3.2
	RS-AMCNN	1057.45±33.2	94.8±8.5

TABLE X
RUNNING TIME OF DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATA SET

Dataset	Method	Training Time (s)	Test Time (s)
Pavia University	RBF-SVM[27]	0.5±0.1	3.5±0.1
	SAE[32]	82.2±5.3	0.3±0.1
	DBN[35]	147.0±10.6	0.4±0.2
	CNN[50]	371.8±15.3	1.2±0.1
	PPF-CNN[44]	4367.9±29.5	7.2±0.4
	3DCNN[37]	2979.0±12.6	31.4±5.5
	DFFN[43]	181.2±1.0	6.2±0.0
	DR-CNN[47]	1817.3±20.8	142.3±6.3
	RS-AMCNN	1655.5±7.9	129.7±10.5

TABLE XI
RUNNING TIME OF DIFFERENT METHODS ON THE UNIVERSITY OF HOUSTON DATA SET

Dataset	Method	Training Time (s)	Test Time (s)
University of Houston	RBF-SVM[27]	0.4±0.1	0.8±0.1
	SAE[32]	78.9±12.3	0.3±0.1
	DBN[35]	132.6±15.8	0.4±0.1
	CNN[50]	306.0±25.5	0.6±0.1
	PPF-CNN[44]	2101.3±26.0	5.3±0.4
	3DCNN[37]	2878.0±39.6	17.0±0.1
	DFFN[43]	123.4±0.1	2.9±0.1
	DR-CNN[47]	1504.0±28.9	115.3±1.6
	RS-AMCNN	1314.5±13.8	99.9±3.2

information effectively. Among all these methods, RS-AMCNN provides consistently promising performance with different ratios of training samples. Thus, RS-AMCNN is more suitable for HSI data sets with limited number of training samples.

F. Effectiveness Analysis of Each Step in RS-AMCNN

There are three vital components in RS-AMCNN, nonlocal adaptive spatial window search, ConvLSTMs and CNN-based multilayer spatial–spectral feature extraction, and multiple branch attention. To valid the effectiveness of three steps, CNN, CNN with ConvLSTMs (MCNN), MCNN with nonlocal adaptive spatial window search (RS-MCNN), and RS-MCNN with multiple branch attention (RS-AMCNN) are used for comparison. Besides, to valid the effectiveness of multiscale and multicontextual information in the proposed method, MCNN with single scale and contextual search (MCNN-SS-SC), MCNN with only multicontextual search (MCNN-MC), and RS-MCNN with only multiscale search (MCNN-MS) are also carried out. Table XII lists the OA results of these methods on three HSI data sets. For CNN and MCNN, the size of spatial window is set to be 23 on the Indian Pines [51], [52], 27 on the Pavia University [37], [53], and 21 on the University of Houston [52], [54].

Compared with CNN, MCNN not only provides joint representation for spatial–spectral structure of HSIs, but also integrates hierarchical features to extract multilayer spatial-spectral features for classification. Compared with MCNN, RS-MCNN achieves 1.4%, 1.8%, and 1.2% improvement in the OA index on three data sets, which shows that adaptive region search is beneficial for classification.

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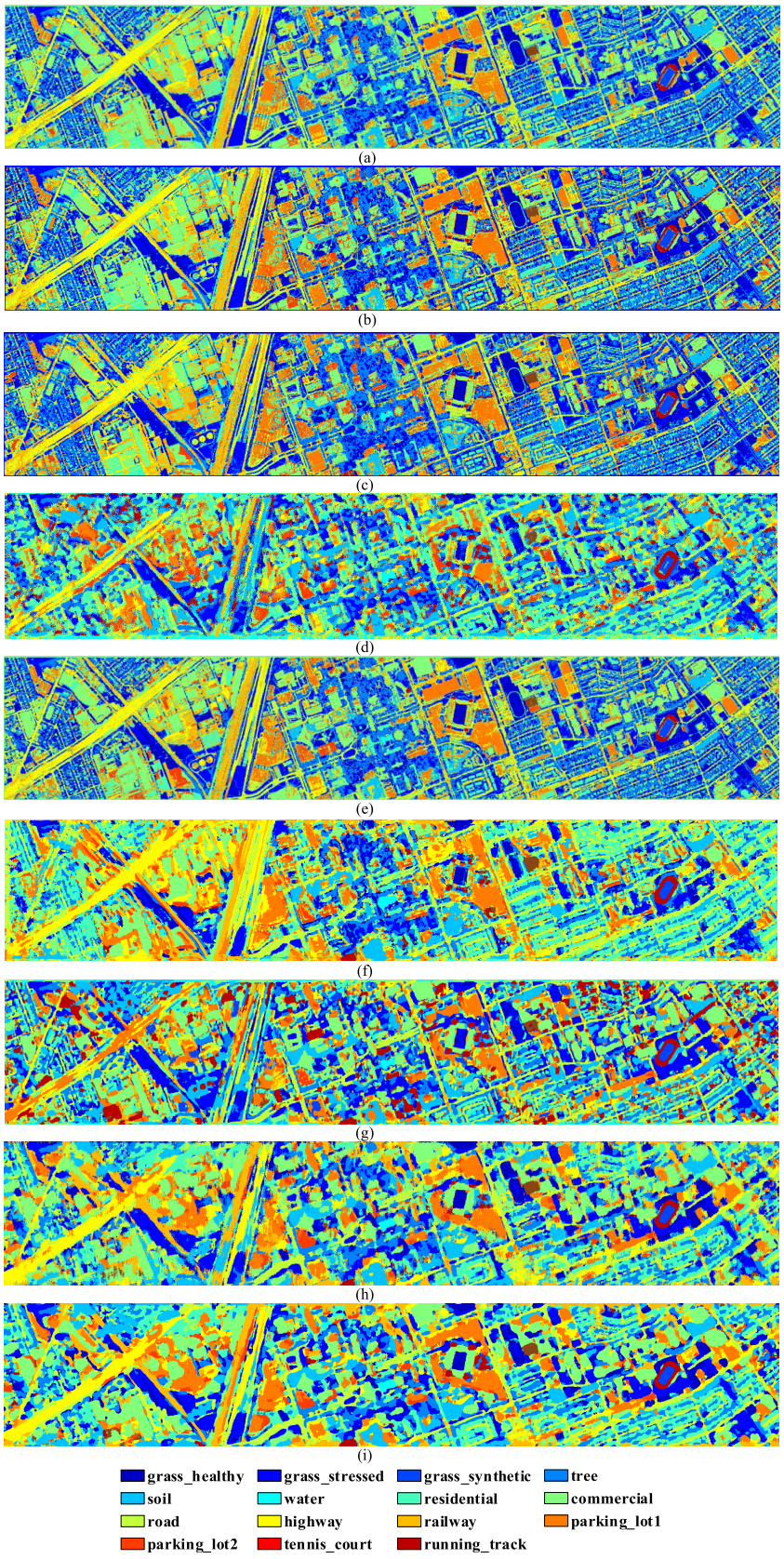


Fig. 10. Classification visual maps of all the nine methods on the University of Houston data set. (a) RBF-SVM. (b) SAE. (c) DBN. (d) CNN. (e) PPF-CNN. (f) 3DCNN. (g) DFFN. (h) DR-CNN. (i) RS-AMCNN.

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TABLE XII
OA RESULTS OF DIFFERENT STEPS OF RS-AMCNN

Method	Indian Pines	Pavia University	University of Houston
CNN	86.1±0.8	94.0±0.7	91.5±0.8
MCNN	96.9±0.7	97.2±0.7	95.9±0.3
MCNN-SS-SC	97.3±0.7	97.9±0.5	96.8±0.6
MCNN-MC	97.4±0.6	98.6±0.4	97.1±0.5
MCNN-MS	97.6±0.5	98.8±0.6	97.3±0.4
RS-MCNN	98.3±0.5	99.0±0.4	98.1±0.3
RS-AMCNN	98.7±0.5	99.5±0.3	98.7±0.2

TABLE XIII
OA RESULTS OF DIFFERENT SIZES OF NONLOCAL CANDIDATE REGIONS IN RS-AMCNN

Indian Pines	Size	32	36	40	44	48
	OA (%)	98.0±0.4	98.5±0.5	98.7±0.5	98.3±0.6	98.2±0.8
Pavia University	Size	40	44	48	52	56
	OA (%)	99.2±0.2	99.3±0.1	99.5±0.1	99.2±0.2	99.2±0.3
University of Houston	Size	46	50	54	58	62
	OA (%)	98.4±0.4	98.5±0.3	98.7±0.2	98.5±0.5	98.4±0.9

TABLE XIV
OA RESULTS OF DIFFERENT JOINT SPATIAL–SPECTRAL FEATURE EXTRACTION METHODS

Method	Indian Pines	Pavia University	University of Houston
MCNN	96.9±0.7	97.2±0.7	95.9±0.3
3DCNN	92.8±0.9	95.2±0.4	93.5±0.5
SSUN	91.8±0.2	93.6±0.3	92.4±0.3

Compared with MCNN, MCNN-SS-SC significantly improves the classification performance using a flexible input. Compared with MCNN-SS-SC, MCNN-MS and MCNN-MC further improve the classification performance by making full use of abundant multiscale information and multicontextual information. RS-MCNN provides more promising classification performance than MCNN-MS and MCNN-MC, which reveals that joint multiscale and multicontextual information is beneficial for HSI classification. Compared with RS-MCNN, RS-AMCNN further elevates the classification performance to a certain extent by dynamically fusing features with different discriminative abilities from multiple branches.

G. Sensitivity Analysis of Free Parameters in RS-AMCNN

In RS-AMCNN, the size of nonlocal candidate region is an important parameter. It is selected by adopting a trial-and-error procedure. This parameter needs to be given before processing, which provides a search space for RS-AMCNN. Here, the size of nonlocal candidate region is investigated to evaluate how the size affects the classification performance.

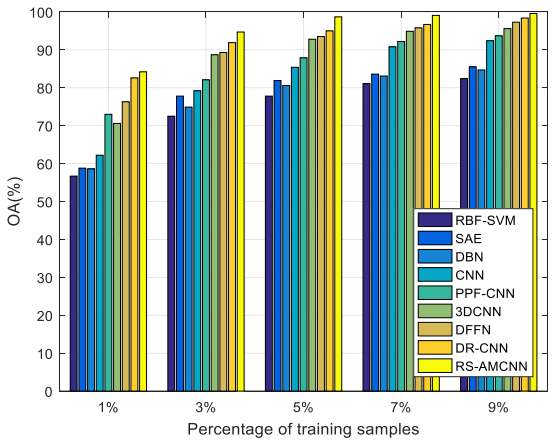
As shown in Table XIII, different sizes of nonlocal candidate region are investigated. The sizes are set to [32], [36], [40], [44], and [48] for Indian Pines, [40], [44], [48], [52], and [56] for Pavia University, and [46], [50], [54], [58], and [62] for University of Houston. It is easy to discover that the most appropriate size is around 40 for the Indian Pines data set, 48 for the Pavia University data set, and 54 for the University of Houston data set. Since the Indian Pines, Pavia University and University of Houston data sets are varying in the spatial resolution, their optimal sizes of nonlocal candidate regions are different. Compared with Indian Pines data set, other data sets are higher in the spatial resolution, which have richer spatial information for classification. Accordingly, the optimal sizes for the Pavia University and University of Houston are larger than that for the Indian Pines. Finally, the size of nonlocal candidate region is set to 40 for the Indian Pines data set, 48 for the Pavia University data set, and 54 for the University of Houston data set.

H. Effectiveness Analysis of Joint Spatial–Spectral Feature Extraction in RS-AMCNN

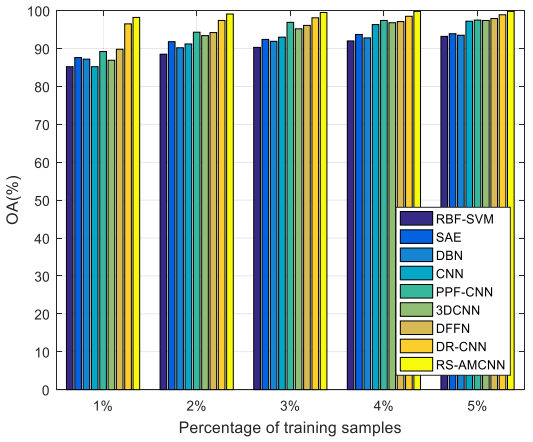
To validate the effectiveness of joint spatial–spectral feature extraction of the proposed method, two comparison experiments are carried out, 3DCNN [37] and SSUN [38]. SSUN is RNN-based spatial–spectral feature extraction method, which uses LSTM to extract spectral features and CNN to extract spatial features. In the proposed method, CNN with ConvLSTMs (MCNN) is retained to validate the effectiveness of joint spatial–spectral feature extraction.

Table XIV records the OA results and corresponding standard deviations over 30 runs. It can be seen that 3DCNN outperforms SSUN by simultaneously making full use of the HSIs' 3-D data structure instead of processing spectral and spatial features separately. Compared with 3DCNN, MCNN performs better. MCNN not only takes the spatial–spectral unified structure into consideration, but also models the long-term dependencies over different spectral bands.

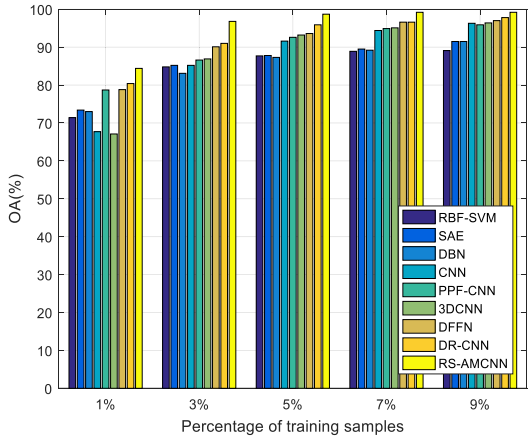
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(a)



(b)



(c)

Fig. 11. OA results of the nine methods with different numbers of training samples on (a) Indian Pines, (b) Pavia University, and (c) University of Houston.

IV. Conclusion

In this article, a novel RS-AMCNN method has been developed to deal with the HSIs with various land-cover distributions. RS-AMCNN transforms conventional single or several fixed spatial window input to a more flexible classification task-driven search manner. In the feature extraction, ConvLSTMs are hierarchically merged into multibranch CNN from shallow to deep layers, which not only exploits spatial–spectral information also extracts multilayer features from detail to abstraction. Moreover, different discriminative abilities of multiple branches are taken into account in RS-AMCNN. In RS-AMCNN, a branch attention mechanism is devised, which impels RS-AMCNN to emphasize more discriminative branches and suppress less useful ones. It can dynamically fuse features from multiple branches. These designs are working in a uniform framework to elevate the classification performance of HSIs. Experimental results further invalid the effectiveness of RS-AMCNN for HSI classification. In the near future, we will consider more effective architectures of joint spatial–spectral feature extraction and other deep learning models for HSI classification.

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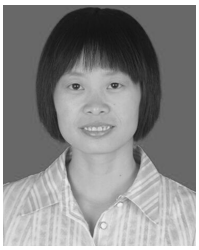
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